

SHREYA BHATTACHARYA

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PROFILE

Electronics and Communication Engineering graduate with a strong foundation in embedded systems, digital electronics, and C++. Hands-on experience in IoT-based projects, microcontroller interfacing, and hardware debugging. Quick learner with strong analytical and problem-solving skills, eager to contribute to hardware and semiconductor engineering.

EDUCATION

B.Tech Electronics and Communication Engineering , Vellore Institute Of Technology CGPA: 8.52	2022 – 2026 <i>Bhopal, M.P.</i>
Higher Secondary – PCM , St. Xavier's High School 10th: 89.8% 12th: 84.16%	2019 – 2021 <i>Hardoi, U.P.</i>

SKILLS

Languages	C/C++, Python (Basic), MATLAB
Tools	Git, GitHub, Docker(basic) VS Code, Raspberry Pi, Arduino
Concepts	SQL, Data Structures & Algorithms, OOP, DBMS, Excel

CERTIFICATIONS

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| • Microsoft Excel Beginners to Advance – Skill Course | <i>Aug 2026</i> |
| • MATLAB Onramp & Simulink Onramp – MathWorks | <i>Aug 2024</i> |
| • Computer Vision – Vityarthi | <i>Aug 2024 – Oct 2024</i> |

PROJECTS

Deep Packet Inspection (DPI) Engine *Ongoing*

- Built a C++ DPI engine for PCAP parsing, network protocol inspection, TLS SNI extraction, and application classification.
- Implemented five-tuple flow tracking, rule-based traffic blocking, and multi-threaded packet processing using a producer-consumer architecture.
- Designed a multi-threaded producer-consumer architecture with thread-safe queues to improve packet-processing scalability for larger captures.

Rubik's Cube Solver *Feb 2026*

- Developed a 3×3 Rubik's Cube Solver in C++ using BFS and DFS algorithms.
- Designed a 3D array-based cube representation for efficient state management and move generation.
- Applied Object-Oriented Programming and STL (vector, queue, stack, unordered_map) to build a modular and efficient solver.

Criminal Detection Using Face Recognition *Apr 2025*

- Deployed a real-time face detection surveillance system using OpenCV, Python, and Raspberry Pi with Haar Cascade classifiers.
- Optimized detection accuracy across diverse lighting and environmental conditions for rapid criminal database matching.

EXTRACURRICULAR ACTIVITIES

Technical Lead, VITRONIX Club, VIT Bhopal *Aug 2024 – Aug 2025*

- Led a 25-member technical team, increasing club engagement by 40% through weekly workshops and hands-on coding sessions.

Volunteer, National Symposium, – ANRF *Feb 2025*

- Contributed to event execution and logistics management; developed coordination and technical event planning skills.